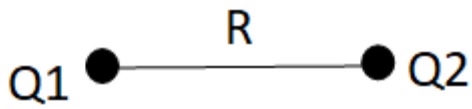
UNIT-I
STATIC ELECTRIC FIELDS

<u>Coulomb's law:</u> Coulomb's law states that the force F between two point charges Q_1 and *Q2* is:
1. Along the line joining them
2. Directly proportional to the product Q_1Q_2 of the charges
3. Inversely proportional to the square of the distance *R* between them



Expressed mathematically as

$$\mathbf{F} \propto \frac{\mathbf{Q_1Q_2}}{\mathbf{R^2}}$$

$$\mathbf{F} = \mathbf{K}\frac{\mathbf{Q_1Q_2}}{\mathbf{R^2}}$$

Where *k* is the proportionality constant. In SI units, charges Q1 and Q2 are in coulombs (C), The distance R is in meters (m), and the force F is in newtons (N) so that $\boldsymbol{K} = \frac{\mathbf{1}}{\mathbf{4\pi\varepsilon_o}}$
The constant $\boldsymbol{\varepsilon_o}$ is known as the permittivity of free space and $\boldsymbol{\varepsilon_o}$ = 8.854 X 10^{-12} F/m.

Therefore K=9X 10^9 m/F.

<u>Vector form of Coulomb's law :</u>

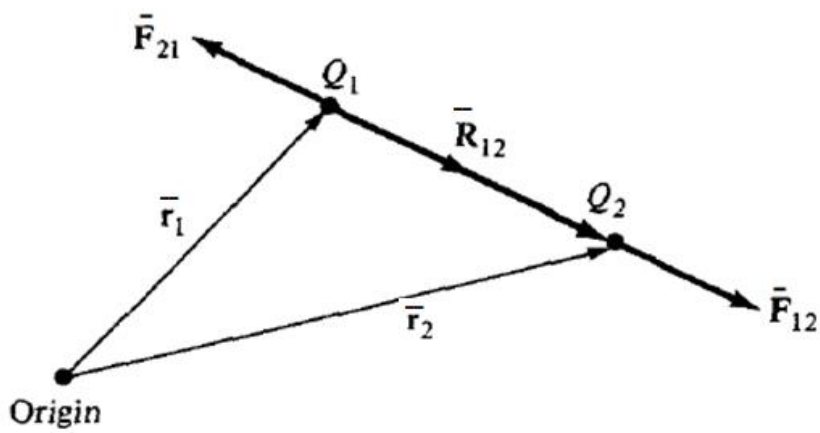


Fig. Vector form of Coulomb's law

If point charges Q_1 and Q_2 are located at points having position vectors r_1 and r_2, then the force $\bar{F}_{12}$ on Q_2 due to Q_1 is given by

$$\bar{\mathbf{F}}_{\mathbf{12}} = \frac{\mathbf{Q_1Q_2}}{\mathbf{4\pi\varepsilon_o R^2}}\bar{\mathbf{a}}_{\mathbf{R_{12}}}$$

Where $\bar{R}_{12} = \bar{r}_2 - \bar{r}_1$

$$R = |\bar{R}_{12}| = |\bar{r}_2 - \bar{r}_1|$$

$$\bar{a}_{R_{12}} = \frac{\bar{R}_{12}}{R}$$

$$\bar{F}_{12} = \frac{Q_1 Q_2}{4\pi\varepsilon_o R^3}\bar{R}_{12}$$

$$\bar{F}_{12} = \frac{Q_1 Q_2(\bar{r}_2 - \bar{r}_1)}{4\pi\varepsilon_o|\bar{r}_2 - \bar{r}_1|^3}$$

Similarly the force F_{21}, on Q_1 due to Q2 is given by

$$\bar{F}_{21} = -\bar{F}_{12}$$

Electric field intensity ($\bar{E}$):

The electric field intensity $\bar{E}$ is the force per unit charge when placed in the electric field.

$$\bar{E} = \lim_{q\to 0}\frac{\bar{F}}{q}$$

$$\bar{E} = \frac{\bar{F}}{q}$$

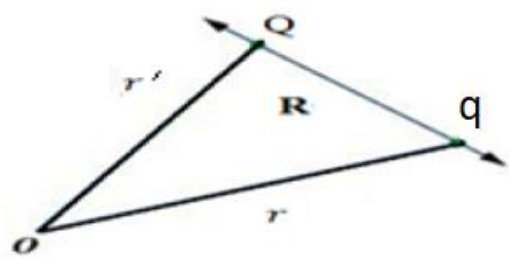


Fig. Electric field intensity

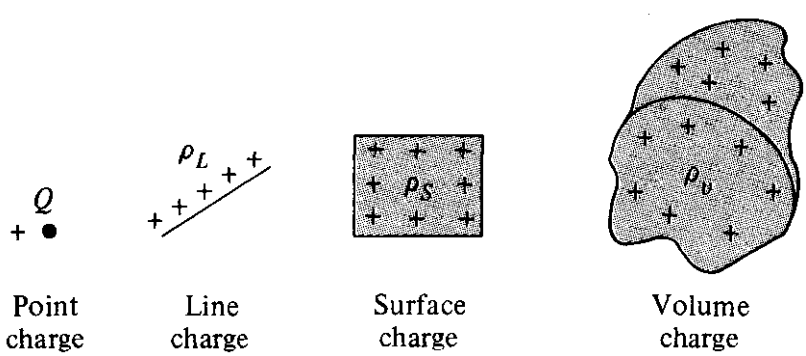
The electric field intensity at point r due to a point charge Q located at r' is given

$$\bar{F} = \frac{qQ}{4\pi\varepsilon_o R^2}\bar{a}_R$$

$$\bar{E} = \frac{Q}{4\pi\varepsilon_o R^2}\bar{a}_R$$

$$\bar{E} = \frac{Q}{4\pi\varepsilon_o R^3}\bar{R} = \frac{Q(\bar{r} - \bar{r}')}{4\pi\varepsilon_o|\bar{r} - \bar{r}'|^3}$$

Charge distributions:



There are four types of charge distributions
1. Point charge
2. Line charge
3. Surface charge
4. Volume charge

<u>Point charge:</u> A small amount of charge occupying infinitesimal volume is called point charge and its units are coulombs.

<u>Line charge:</u> A charge distributed uniformly along a line is called line charge. It is denoted by ρ_L. Its units are C/m.

$$\rho_L = \frac{dQ}{dl}$$
$$dQ = \rho_L dl$$
$$\mathbf{Q} = \int_{\mathbf{L}} \boldsymbol{\rho}_{\mathbf{L}} \mathbf{dl}$$

The electric filed intensity due to line charge is

$$\bar{\mathbf{E}} = \int \frac{\boldsymbol{\rho}_{\mathbf{L}} \mathbf{dl}}{\mathbf{4\pi\varepsilon_o R^2}} \bar{\mathbf{a}}_{\mathbf{R}}$$

<u>Surface charge:</u> A charge distributed uniformly over a surface is called line charge. It is denoted by ρ_S. Its units are C/m^2.

$$\rho_S = \frac{dQ}{dS}$$
$$dQ = \rho_{\mathbf{S}} d\mathbf{S}$$
$$\mathbf{Q} = \int_{\mathbf{S}} \boldsymbol{\rho}_{\mathbf{S}} \mathbf{dS}$$

The electric filed intensity due to line charge is

$$\bar{\mathbf{E}} = \int \frac{\boldsymbol{\rho}_{\mathbf{S}} \mathbf{dS}}{\mathbf{4\pi\varepsilon_o R^2}} \bar{\mathbf{a}}_{\mathbf{R}}$$

<u>Volume charge:</u> A charge distributed uniformly along over a volume is called line charge. It is denoted by ρ_V. Its units are C/m^3.

$$\rho_{\mathrm{V}} = \frac{dQ}{d\mathrm{V}}$$
$$dQ = \rho_{\mathrm{V}} d\mathrm{V}$$
$$\mathbf{Q} = \int_{\mathrm{V}} \boldsymbol{\rho}_{\mathrm{V}} \mathbf{d}\mathrm{V}$$

The electric filed intensity due to line charge is

$$\bar{\mathbf{E}} = \int \frac{\boldsymbol{\rho}_{\mathrm{V}} \mathbf{d}\mathrm{V}}{\mathbf{4\pi\varepsilon_o R^2}} \bar{\mathbf{a}}_{\mathbf{R}}$$

<u>Electric field intensity due to finite Line Charge:</u>

Consider a line charge with uniform charge density ρ_L extending from *A* to *B* along the z-axis as shown in Figure.

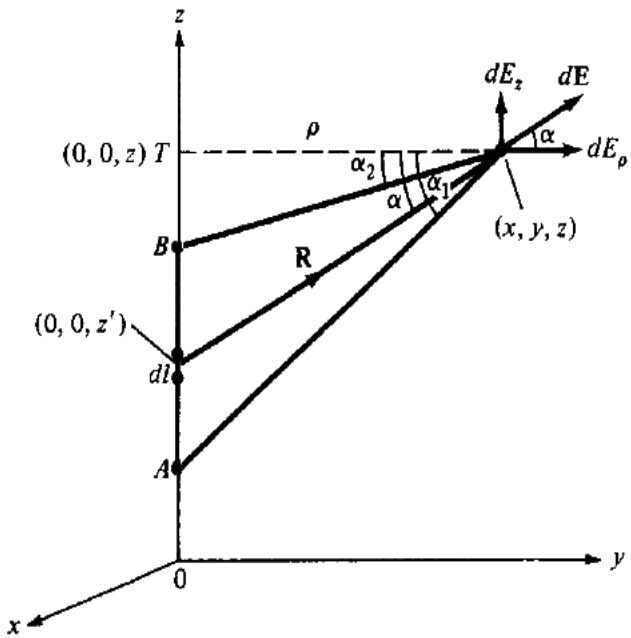


Fig . Electric Filed Intensity due to a line charge

Let us consider differential element dl on the line at z' and the charge element dQ associated with element dl of the line is

$$\mathbf{dQ = \rho_L dl}$$

Hence the total charge Q is

$$\mathbf{Q = \int \rho_L dl}$$

The electric field intensity $\bar{E}$ at an arbitrary point P(x, y, z) can be found using equation

$$\mathbf{\bar{E} = \frac{Q}{4\pi\varepsilon_o R^2}\bar{a}_R}$$

$$\mathbf{\bar{E} = \int \frac{\rho_L dl}{4\pi\varepsilon_o R^2}\bar{a}_R \quad ---(1)}$$

If the dl is located at (0,0,z') then dl=dz'

$$\mathbf{\bar{R} = \rho\bar{a}_\rho + \left(z - z'\right)\bar{a}_z \quad ---(2)}$$

$$\mathbf{R = \sqrt{\rho^2 + \left(z - z'\right)^2}}$$

$$\mathbf{\bar{a}_R = \frac{\bar{R}}{R} = \frac{\rho\bar{a}_\rho + \left(z - z'\right)\bar{a}_z}{\sqrt{\rho^2 + \left(z - z'\right)^2}} \quad ---(3)}$$

Substituting these equations in eq(1) then we get

$$\mathbf{\bar{E} = \frac{\rho_L}{4\pi\varepsilon_o}\int_{z_A}^{z_B} \frac{\rho\bar{a}_\rho + \left(z - z'\right)\bar{a}_z}{\left[\rho^2 + \left(z - z'\right)^2\right]^{\frac{3}{2}}} dz'}$$

To evaluate this, it is convenient that we define α , α_1 and α_2 as in Figure

$$z - z' = \rho \tan\alpha$$
$$dz' = -\rho sec^2\alpha \, d\alpha$$
$$\bar{E} = \frac{-\rho_L}{4\pi\varepsilon_o}\int_{\alpha_1}^{\alpha_2} \frac{\rho\bar{a}_\rho + \rho\tan\alpha\,\bar{a}_z}{[\rho^2 + \rho^2\tan^2\alpha]^{\frac{3}{2}}}\,\rho sec^2\alpha\, d\alpha$$
$$\rho^2 + \rho^2\tan^2\alpha = \rho^2 sec^2\alpha$$
$$\bar{E} = \frac{-\rho_L}{4\pi\varepsilon_o}\int_{\alpha_1}^{\alpha_2} \frac{\bar{a}_\rho + \tan\alpha\,\bar{a}_z}{\rho^3 sec^3\alpha}\,\rho^2 sec^2\alpha\, d\alpha$$
$$\bar{E} = \frac{-\rho_L}{4\pi\varepsilon_o}\int_{\alpha_1}^{\alpha_2} \frac{\bar{a}_\rho + \tan\alpha\,\bar{a}_z}{\rho\sec\alpha}\, d\alpha$$
$$\bar{E} = \frac{-\rho_L}{4\pi\varepsilon_o\rho}\int_{\alpha_1}^{\alpha_2} \left[\cos\alpha\,\bar{a}_\rho + \sin\alpha\,\bar{a}_z\right] d\alpha$$
$$\bar{E} = \frac{-\rho_L}{4\pi\varepsilon_o\rho}\left[(\sin\alpha_2 - \sin\alpha_1)\bar{a}_\rho - (\cos\alpha_2 - \cos\alpha_1)\right]\bar{a}_z$$

Thus, the electric field intensity for a finite line charge

$$\bar{E} = \frac{\rho_L}{4\pi\varepsilon_o\rho}\left[-(\sin\alpha_2 - \sin\alpha_1)\bar{a}_\rho + (\cos\alpha_2 - \cos\alpha_1)\right]\bar{a}_z$$

For an infinite line charge α_1=90◦ and α_2=-90◦

$$\boxed{\bar{E} = \frac{\rho_L}{2\pi\varepsilon_0\rho}\bar{a}_\rho}$$

<u>Electric field intensity due to circular ring of Charge:</u>

Let us consider a circular ring of radius *a* carries a uniform charge ρ_L C/m and is placed on the xy-plane with axis the same as the z-axis as shown in fig.

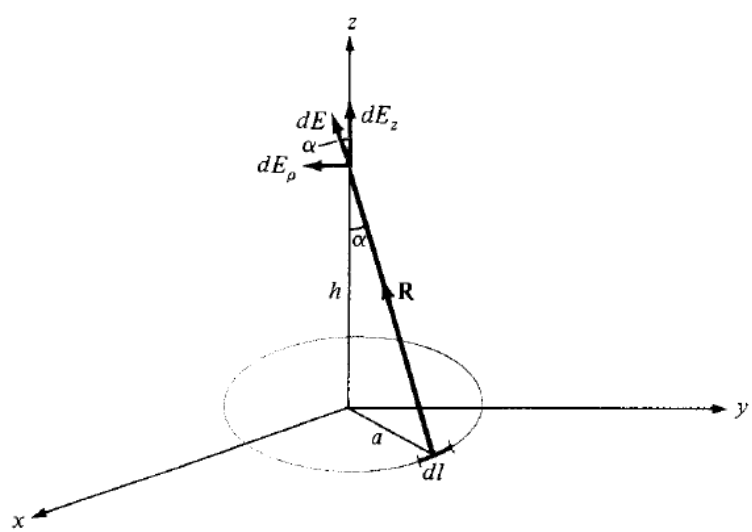


Fig. the circular ring of Charge

The electric field intensity at pont p(0,0,h) is given by

$$\bar{E} = \int \frac{\rho_L dl}{4\pi\varepsilon_o R^2}\bar{a}_R \quad ---(1)$$

In this case, $\mathbf{dl = ad\phi}$

$$\bar{\mathbf{R}} = -\mathbf{a}\bar{\mathbf{a}}_\rho + \mathbf{h}\bar{\mathbf{a}}_z$$
$$\mathbf{R} = \sqrt{\mathbf{a}^2 + \mathbf{h}^2}$$
$$\bar{a}_R = \frac{\bar{R}}{R} = \frac{-\mathbf{a}\bar{\mathbf{a}}_\rho + \mathbf{h}\bar{\mathbf{a}}_z}{\sqrt{\mathbf{a}^2 + \mathbf{h}^2}}$$

Substitute the above equations in equation (1)

$$\bar{\mathbf{E}} = \frac{\rho_L}{4\pi\varepsilon_o}\int_0^{2\pi} \frac{\left(-\mathbf{a}\bar{\mathbf{a}}_\rho + \mathbf{h}\bar{\mathbf{a}}_z\right)}{[\mathbf{a}^2 + \mathbf{h}^2]^{\frac{3}{2}}}\, a\mathbf{d\phi}$$

Due to symmetry all radial components are cancelled to each other.

$$\bar{\mathbf{E}} = \frac{\rho_L}{4\pi\varepsilon_o}\int_0^{2\pi} \frac{\mathbf{h}\bar{\mathbf{a}}_z}{[\mathbf{a}^2 + \mathbf{h}^2]^{\frac{3}{2}}}\, a\mathbf{d\phi}$$
$$\bar{\mathbf{E}} = \frac{\rho_L \mathbf{a}\mathbf{h}\bar{\mathbf{a}}_z}{4\pi\varepsilon_o[\mathbf{a}^2 + \mathbf{h}^2]^{\frac{3}{2}}}\int_0^{2\pi} \mathbf{d\phi}$$

$$\bar{\mathbf{E}} = \frac{\rho_L a\mathbf{h}\bar{\mathbf{a}}_z}{4\pi\varepsilon_o[\mathbf{a}^2 + \mathbf{h}^2]^{\frac{3}{2}}}\, 2\pi$$
$$\bar{\mathbf{E}} = \frac{\rho_L a\mathbf{h}}{2\varepsilon_o[\mathbf{a}^2 + \mathbf{h}^2]^{\frac{3}{2}}}\, \bar{\mathbf{a}}_z$$

<u>Electric field intensity due to an infinite charge sheet:</u>

Consider an infinite charge sheet in the xy-plane with uniform charge density ρ_s as shown in fig.

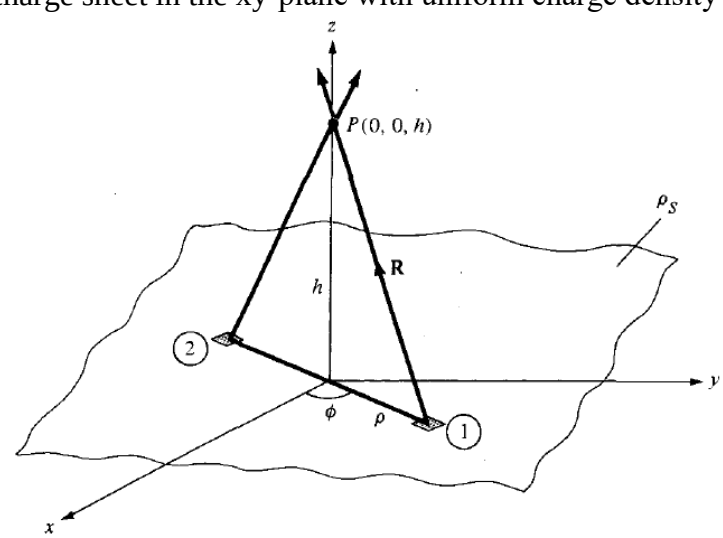


Fig. an infinite charge sheet

The charge associated with the differential area dS is $dQ = \rho_s dS$

The total charge on the sheet

$$Q = \int_S \rho_s dS$$

The electric field intensity at pont p(0,0,h) is given by

$$\bar{\mathbf{E}} = \int_S \frac{\rho_s dS}{4\pi\varepsilon_o R^2} \bar{a}_R \quad ----(1)$$

In this case,

$$\bar{\mathbf{R}} = -\rho\bar{a}_\rho + h\bar{a}_z$$
$$R = \sqrt{\rho^2 + h^2}$$
$$\bar{a}_R = \frac{\bar{R}}{R} = \frac{-\rho\bar{a}_\rho + h\bar{a}_z}{\sqrt{\rho^2 + h^2}}$$

Substitute the above equations in equation (1)

$$\bar{\mathbf{E}} = \frac{\rho_S}{4\pi\varepsilon_o} \int_S \frac{\left(-\rho\bar{a}_\rho + h\bar{a}_z\right)}{[\rho^2 + h^2]^{\frac{3}{2}}} dS$$

Due to symmetry all radial components are cancelled to each other.

$$\bar{\mathbf{E}} = \frac{\rho_S}{4\pi\varepsilon_o} \int_S \frac{h\bar{a}_z}{[\rho^2 + h^2]^{\frac{3}{2}}} dS$$

Where

$$dS = \rho d\rho d\phi$$

$$\bar{\mathbf{E}} = \frac{\rho_S h}{4\pi\varepsilon_o} \int_0^{2\pi} \int_{\rho=0}^{\infty} \frac{\rho d\rho d\phi}{[\rho^2 + h^2]^{\frac{3}{2}}} \bar{a}_z$$
$$\bar{\mathbf{E}} = \frac{\rho_S h}{4\pi\varepsilon_o} \int_0^{2\pi} d\phi \int_{\rho=0}^{\infty} \frac{\rho d\rho}{[\rho^2 + h^2]^{\frac{3}{2}}} \bar{a}_z$$

Let $t^2 = \rho^2 + h^2$ **then** $tdt = \rho d\rho$

$$\bar{\mathbf{E}} = \frac{\rho_S h}{4\pi\varepsilon_o} 2\pi \int_{t=h}^{\infty} \frac{tdt}{t^3} \bar{a}_z$$

$$\bar{\mathbf{E}} = \frac{\rho_S h}{2\varepsilon_o} \int_{t=h}^{\infty} \frac{dt}{t^2} \bar{a}_z$$
$$\bar{\mathbf{E}} = \frac{\rho_S h}{2\varepsilon_o} \left[\frac{-1}{t}\right]_h^{\infty} \bar{a}_z = \frac{\rho_S h}{2\varepsilon_o} \frac{1}{h} \bar{a}_z$$

The electric field intensity due to infinite charge sheet if point p is on positive Z-axis is

$$\bar{\mathbf{E}} = \frac{\rho_S}{2\varepsilon_o} \bar{a}_z$$

The electric field intensity due to infinite charge sheet if point p is on negative Z-axis is

$$\bar{\mathbf{E}} = -\frac{\rho_S}{2\varepsilon_o} \bar{a}_z$$

Electric Flux Density:
Electric field intensity depends on the medium in which the charges are placed.
To avoid this electric flux density is introduced.
Electric Flux(**ψ**) :
The total number of lines of forces in any particular electric field is known as "electric flux".
It is denoted by **ψ.**
The unit of electric flux is Coulombs(C).
The electric flux displaces charge from one place to other place.
Electric Flux Density(D):
The total electric flux per unit area is defined as the electric flux density.

$$\boxed{\mathbf{D = \frac{d\psi}{ds}\ coulombs/m^2}}$$

$$\boxed{\mathbf{d\psi = Dds}}$$

$$\boxed{\mathbf{\psi = \int_s \bar{D}.\overline{ds}}}$$

$$\boxed{\mathbf{\bar{D} = \varepsilon_0 \bar{E}}}$$

$$\boxed{\mathbf{\bar{D} = \frac{Q}{4\pi R^2}\bar{a}_R}}$$

Gauss's law:
Gauss's law states that the total electric flux ψ through any closed surface is equal to the total charge enclosed by that surface.

$$\boxed{\mathbf{\psi = Q}} \quad \mathbf{---(1)}$$

$$\text{Total flux } \mathbf{\psi = \oint_s \bar{D}\cdot\overline{ds}} \quad \mathbf{---(2)}$$

$$\text{Total charge enclosed } \mathbf{Q = \int_v \rho_v dv} \quad \mathbf{---(3)}$$

Substitute the Eq.(2) and Eq.(3) in equation (1), we get

$$\boxed{\mathbf{\oint_s \bar{D}\cdot\overline{ds} = \int_v \rho_v dv}} \quad \mathbf{----(4)}$$

This equation is called integral form of Gauss's law
By applying divergence theorem to the first term in the above equation

$$\mathbf{\oint_s \bar{D}\cdot\overline{ds} = \int_v \nabla\cdot\bar{D}dv}$$

$$\mathbf{\int_v \nabla\cdot\bar{D}dv = \int_v \rho_v dv}$$

$$\boxed{\mathbf{\nabla\cdot\bar{D} = \rho_v}}$$

The above equation is called point form of gauss's law
Applications of Gauss's law : Gauss's law provides an easy means of finding E or D for symmetrical charge distributions such as a point charge, an infinite line charge, an infinite cylindrical surface charge, and a spherical charge .

The procedure for finding **E** or **D** using Gauss's law:

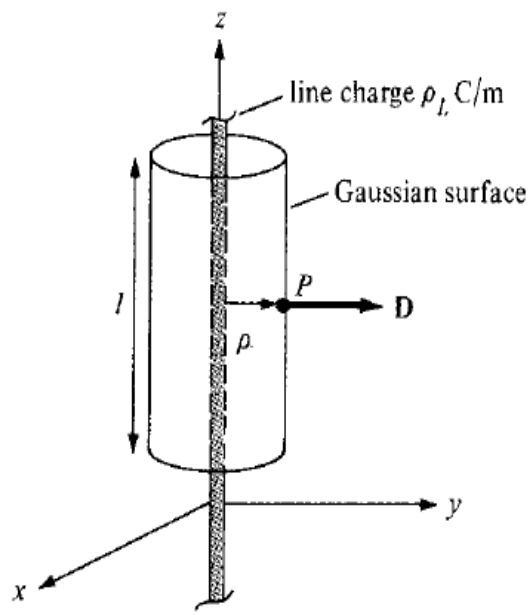
1. Verify whether charge symmetry exists or not.
2. Construct closed surface (Gaussian surface) around the charge distribution.
3. The surface is chosen such that **D** is either normal or tangential to the Gaussian surface.
   (a).When **D** is normal to the surface, **D • dS** = D dS because **D** is constant on the surface.
   (b).When **D** is tangential to the surface, **D • dS** = 0.
4. Use the $\oint_s \overline{\mathbf{D}} \cdot \overline{\mathbf{ds}} = \mathbf{Q}$ to find **E** or **D.**

Limitations of Gauss's law:

- It is useful only for symmetric charge distribution.
- It can applicable only for closed surfaces.
- It can applicable only if the surface under consideration encloses the entire charge.

Electric field intensity due to Infinite Line Charge:

Let us consider the infinite line of uniform charge ρ_L C/m lies along the z-axis.



- Fig. Fig. Gaussian surface about an infinite line charge

To determine D at a point P, construct the cylindrical surface with radius ρ around infinite line charge as Gaussian surface.

From Gauss's law we know that $\oint_s \overline{\mathbf{D}} \cdot \overline{\mathbf{ds}} = \mathbf{Q}$

$$\oint_s \mathbf{D}\ \mathbf{dS} = \mathbf{Q}$$

$$\mathbf{D} \oint_s \mathbf{dS} = \mathbf{Q}$$

Where $\oint_s \mathbf{dS} = \mathbf{2\pi\rho l}$ is the surface area of the Gaussian surface.

$\oint_s \overline{\mathbf{D}} \cdot \overline{\mathbf{ds}}$ evaluated on the top and bottom surfaces of the cylinder is zero, since D is tangential to the top and bottom surfaces.

$$\mathbf{D(2\pi\rho l)} = \boldsymbol{\rho}_\mathbf{L}\ \mathbf{l}$$

$$\mathbf{D} = \frac{\rho_L}{2\pi\rho}$$

The electric flux density is

$$\bar{\mathbf{D}} = \frac{\rho_L}{2\pi\rho}\bar{\mathbf{a}}_\rho$$

Electric field intensity due to infinite line charge is

$$\bar{\mathbf{E}} = \frac{\rho_L}{2\pi\varepsilon_0\rho}\bar{\mathbf{a}}_\rho$$

<u>Electric field intensity due to Infinite Sheet of Charge:</u>

Consider the infinite sheet of uniform charge ρ_s C/m^2 lying on the $z = 0$ plane as shown in fig.

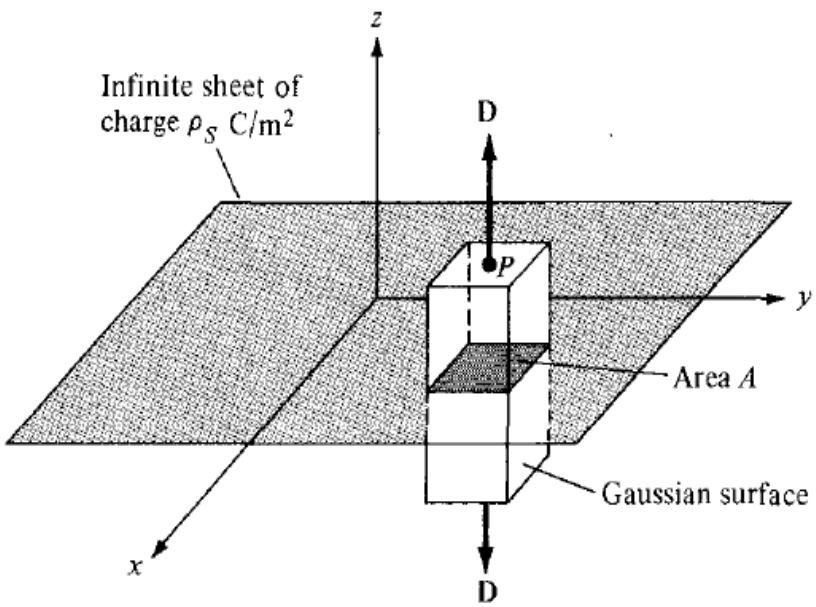


Fig. An infinite sheet of charge.

To determine E at a point P, construct rectangular box as shown in fig.

From Gauss's law we know that

$$\oint_s \bar{\mathbf{D}} \cdot \overline{\mathbf{ds}} = \mathbf{Q}$$

$$\oint_{\mathbf{top}} \bar{\mathbf{D}} \cdot \overline{\mathbf{ds}} + \oint_{\mathbf{bottom}} \bar{\mathbf{D}} \cdot \overline{\mathbf{ds}} = \mathbf{Q}$$

Note that $\bar{\mathbf{D}} \cdot \overline{\mathbf{ds}}$ evaluated on the sides of the box is zero because D is tangential to the sides.

$$\oint_{\mathbf{top}} \mathbf{DdS} + \oint_{\mathbf{bottom}} \mathbf{DdS} = \mathbf{Q}$$

$$\mathbf{D}\left(\oint_{\mathbf{top}} \mathbf{dS} + \oint_{\mathbf{bottom}} \mathbf{dS}\right) = \mathbf{Q}$$

If the top and bottom area of the box each has area A, then above equation becomes

$$\mathbf{D}(\mathbf{A} + \mathbf{A}) = \rho_S\mathbf{A}$$

$$\mathbf{D2A} = \rho_S\mathbf{A}$$

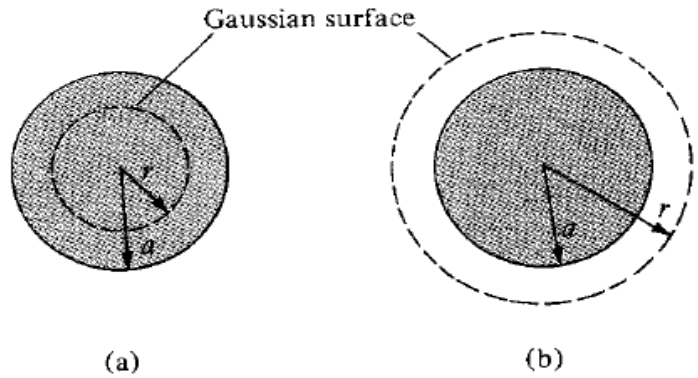
$$\mathbf{D} = \frac{\rho_S}{2}$$
$$\bar{\mathbf{D}} = \frac{\rho_S}{2}\bar{a}_Z$$

Electric field intensity due to infinite sheet charge is

$$\bar{\mathbf{E}} = \frac{\rho_S}{2\varepsilon_0}\bar{a}_Z$$

<u>Electric field intensity due to Uniformly Charged Sphere:</u>

Consider a sphere of radius *a* with a uniform charge ρ_v C/m^3.



- Fig. Gaussian surface for a uniformly charged sphere when: (a) r ≤ a and (b) r ≥ a.

To determine E at a point P, Construct the spherical surface of radius r around the charged sphere as shown in fig.

- Consider two cases (a) r ≤ a and (b) r ≥ a.

<u>Case (1):</u> r ≤ a

From Gauss's law we know that

$$\oint_s \mathbf{DdS} = \mathbf{Q}$$
$$\mathbf{D}\oint_s \mathbf{dS} = \mathbf{Q}$$
$$\mathbf{D4\pi r^2} = \mathbf{Q}$$

The surface area of the sphere is $4\pi r^2$.

For r ≤ a, the total charge enclosed by the spherical surface of radius r, as shown in Fig (a), is

$$\mathbf{Q} = \rho_v \frac{4}{3}\pi r^3$$

Therefore

$$\mathbf{D4\pi r^2} = \rho_v \frac{4}{3}\pi r^3$$
$$\mathbf{D} = \frac{\rho_v}{3}\mathbf{r}$$
$$\bar{\mathbf{D}} = \frac{\mathbf{r}}{3}\rho_v \bar{a}_r$$

Electric field intensity inside the charged sphere is

$$\bar{\mathbf{E}} = \frac{\mathbf{r}}{3\varepsilon_0}\rho_v\, \bar{a}_r$$

Case (2): $r \geq a$

From Gauss's law we know that

$$\oint_s \mathbf{DdS} = \mathbf{Q}$$

$$\mathbf{D}\oint_s \mathbf{dS} = \mathbf{Q}$$

$$\mathbf{D4\pi r^2} = \mathbf{Q}$$

The surface area of the sphere is $4\pi r^2$.

In this case the charge enclosed by the surface is the entire charge, that is,

$$\mathbf{Q} = \rho_v \frac{4}{3}\pi a^3$$

Therefore

$$\mathbf{D4\pi r^2} = \rho_v \frac{4}{3}\pi a^3$$

$$\mathbf{D} = \frac{a^3}{3r^2}\rho_v$$

$$\bar{\mathbf{D}} = \frac{a^3}{3r^2}\rho_v\,\bar{a}_r$$

Electric field intensity inside the charged sphere is

$$\bar{\mathbf{E}} = \frac{a^3}{3\varepsilon_0 r^2}\rho_v\,\bar{a}_r$$

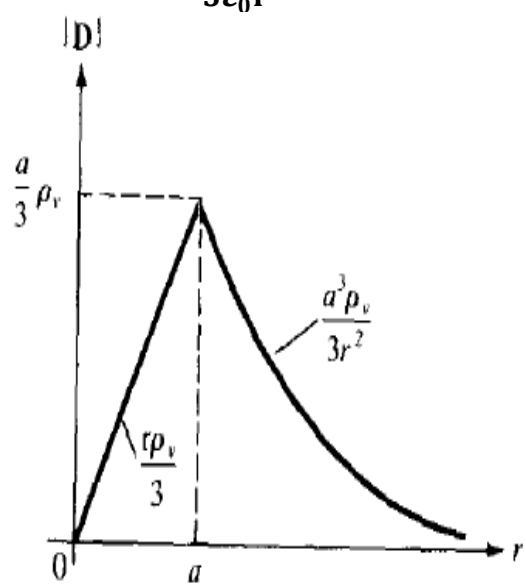


Fig. Sketch of |D| against r for a uniformly charged sphere.

The total work done in moving a point charge from A to B is

$$\mathbf{W} = -\int_A^B \bar{\mathbf{F}} \cdot \bar{\mathbf{dl}}$$

The negative sign indicates that the work is done against to the field.

The force

$$\bar{\mathbf{F}} = \mathbf{Q}\bar{\mathbf{E}}$$

Thus the total work done, in moving Q from A to B is

$$\mathbf{W = -Q\int_A^B \bar{E}\cdot\bar{dl}}$$

The potential difference between points A and B is defined as the work done per unit charge and is given by

$$\mathbf{V_{AB} = -\int_A^B \bar{E}\cdot\bar{dl}}$$

The potential at any point due to a point charge Q located at the origin is

$$\boxed{\mathbf{V = \frac{Q}{4\pi\varepsilon_0 r}}}$$

$$\mathbf{V = \int_L \frac{\rho_L dl}{4\pi\varepsilon_0 r}} \quad \mathbf{(Line\ charge)}$$

$$\mathbf{V = \int_s \frac{\rho_s dS}{4\pi\varepsilon_0 r}} \quad \mathbf{(Surface\ charge)}$$

$$\mathbf{V = \int_v \frac{\rho_v dv}{4\pi\varepsilon_0 r}} \quad \mathbf{(Volume\ charge)}$$

<u>Conservative nature of electrostatic field :</u>

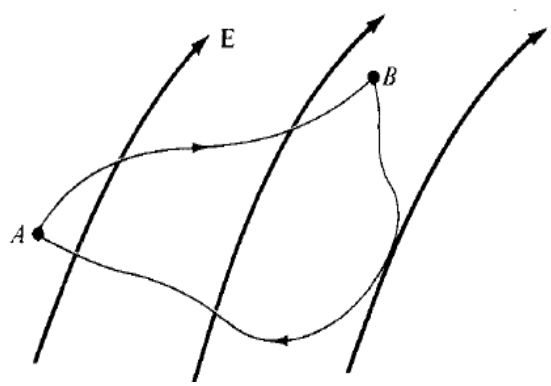


Fig. Conservative nature of an electrostatic field.

The potential difference between points A and B is independent of the path taken. Hence,

$$\mathbf{V_{AB} = -V_{BA}}$$

That is

$$\mathbf{V_{AB} + V_{BA} = 0}$$

$$\mathbf{\oint \bar{E}\cdot\bar{dl} = 0} \quad \mathbf{---(1)}$$

Eq(1) shows that the line integral of E along a closed path as shown in Fig must be zero.

Applying Stokes's theorem to eq. (1) gives

$$\mathbf{\oint \bar{E}\cdot\bar{dl} = \int_s (\nabla\times\bar{E})\cdot\overline{ds} = 0}$$

Or

$$\nabla \times \bar{E} = 0 \qquad ---(2)$$

A field whose line integral does not depend on the path of integration is called conservative field, hence Electrostatic field is conservative field. Eq(2) is also called Maxwell's 2nd equation.

Realation between E and V:

From the definition of potential,

$$V = -\int \bar{E} \cdot \bar{dl} \qquad ---(1)$$

Differentiate Eq.(1) with respect to l, we get

$$dV = -\bar{E} \cdot \bar{dl}$$

Where

$$\bar{E} = E_x\bar{a}_x + E_y\bar{a}_y + E_z\bar{a}_z$$
$$\bar{dl} = dx\,\bar{a}_x + dy\,\bar{a}_y + dz\,\bar{a}_z$$

Therefore

$$dV = -\left(E_x\bar{a}_x + E_y\bar{a}_y + E_z\bar{a}_z\right) \cdot \left(dx\,\bar{a}_x + dy\,\bar{a}_y + dz\,\bar{a}_z\right)$$
$$dV = -E_x dx - E_y dy - E_z dz \qquad ---(1)$$

But from calculus of multiple variables

$$dV = \frac{\partial V}{\partial x}dx + \frac{\partial V}{\partial y}dz + \frac{\partial V}{\partial z}dz \qquad ---(2)$$

Comparing the two expressions for dV, we obtain

$$E_x = -\frac{\partial V}{\partial x}, \qquad E_y = -\frac{\partial V}{\partial y}, and\ E_z = -\frac{\partial V}{\partial z}$$

Thus

$$\bar{E} = E_x\bar{a}_x + E_y\bar{a}_y + E_z\bar{a}_z$$
$$\bar{E} = -\frac{\partial V}{\partial x}\bar{a}_x - \frac{\partial V}{\partial y}\bar{a}_y - \frac{\partial V}{\partial z}\bar{a}_z$$

$$\boxed{\bar{E} = -\nabla V}$$

ELECTRIC DIPOLE :

An electric dipole is formed when two point charges of equal magnitude but opposite sign are separated by a small distance.

The potential at point P(r,θ,ϕ) is given by

$$V = \frac{P\cos\theta}{4\pi\varepsilon_0 r^2}$$

Where the dipole moment P=Qd

The electric filed intensity of dipole is

$$\bar{E} = \frac{P}{4\pi\varepsilon_0 r^3}\left[2\cos\theta\ \bar{a}_r + \sin\theta\ \bar{a}_\theta\right]$$

An equipotential surface:

Any surface on which the potential is the same throughout is known as an equipotential surface. No work is done in moving a charge from one point to another along an equipotential surface and hence

$$\int \bar{E} \cdot \bar{dl} = 0$$

ENERGY DENSITY IN ELECTROSTATIC FIELDS:

Consider an empty space where there is no electric field at all.

Let us move three point charges Q_1,Q_2 and Q_3 from infinity to an empty space as shown in Fig.

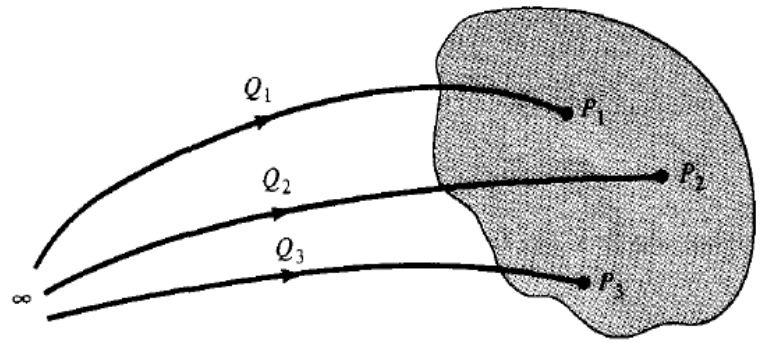


Fig. Assembling of charges.

No work is required to move Q_1 from infinity to P_l because there is no electric field initially [$W_1 = 0$]. The work done in moving Q_2 from infinity to P_2 is equal to the product of Q_2 and the potential V_{21} a t P2 due to Q_1. Similarly, the work done in moving Q_3 to P_3 is equal to Q_3(V32 + V_{31}), where V_{32} and V_{31} are the potentials at P_3 due to Q_2 and Q1 respectively. Hence the total work done in positioning the three charges is

$$\mathbf{W_E = W_1 + W_2 + W_3}$$

$$\mathbf{W_E = 0 + Q_2V_{21} + Q_3(V_{31} + V_{32})} \quad \mathbf{---(1)}$$

If the charges were moved in reverse order,

$$\mathbf{W_E = W_3 + W_2 + W_1}$$

$$\mathbf{W_E = 0 + Q_2V_{23} + Q_1(V_{12} + V_{13})} \quad \mathbf{---(2)}$$

Where V_{23} is the potential at P_2 due to Q_3, V_{l2} and V_{l3} are, respectively, the potentials at P_1 due to Q_2 and Q_3.

Adding eqs. (1) and (2) gives

$$\mathbf{2W_E = Q_1(V_{12} + V_{13}) + Q_2(V_{21} + V_{23}) + Q_3(V_{31} + V_{32})}$$

$$\mathbf{2W_E = Q_1V_1 + Q_2V_2 + Q_3V_3}$$

$$\mathbf{W_E = \frac{1}{2}(Q_1V_1 + Q_2V_2 + Q_3V_3)} \quad \mathbf{---(3)}$$

Where V_1,V_2, and V_3 are total potentials at P_1, P_2, and P_3, respectively.

In general, if there are n point charges, eq. (3) becomes

$$\boxed{\mathbf{W_E = \frac{1}{2}\sum_{k=1}^{n} Q_kV_k \quad (In\ joules)}} \quad \mathbf{---(4)}$$

If instead of point charges, the region has a continuous charge distribution, the summation in eq. (4) becomes integration; that is,

$$\mathbf{W_E = \frac{1}{2}\int \rho_L V dl \ \ (Line\ charge)} \quad \mathbf{---(5)}$$

$$\mathbf{W_E = \frac{1}{2}\int \rho_s V ds \ \ (Surface\ charge)} \quad \mathbf{---(6)}$$

$$\mathbf{W_E = \frac{1}{2}\int \rho_v V dv \ \ (Volume\ charge)} \quad \mathbf{----(7)}$$

We know that $\boldsymbol{\rho_V = \nabla \cdot \bar{D}}$, then equ(7) becomes

$$\mathbf{W_E = \frac{1}{2}\int (\nabla \cdot \bar{D})V dv \quad ---(8)}$$

From vector identity

$$\mathbf{\nabla \cdot V\bar{D} = \bar{D} \cdot \nabla V + V(\nabla \cdot \bar{D})}$$

$$\mathbf{(\nabla \cdot \bar{D})V = \nabla \cdot V\bar{D} - \bar{D} \cdot \nabla V \quad ---(9)}$$

Substitute equ(9) in equ(8) then

$$\mathbf{W_E = \frac{1}{2}\int_v (\nabla \cdot V\bar{D})\, dv - \frac{1}{2}\int_v (\bar{D} \cdot \nabla V) dv}$$

By applying divergence theorem to the first term on the right-hand side of this equation, we Have

$$\mathbf{W_E = \frac{1}{2}\oint_s (V\bar{D}) ds - \frac{1}{2}\int_v (\bar{D} \cdot \nabla V) dv \quad ---(10)}$$

The first term VD in the eq. (10) must vary at least as $1/r^3$ while dS varies as r^2. Consequently, the first integral in eq. (10) must tend to zero as the surface S becomes large. Hence, eq. (10) reduces to

$$\mathbf{W_E = -\frac{1}{2}\int_v (\bar{D} \cdot \nabla V) dv}$$

We know that $\boldsymbol{\bar{E} = -\nabla V}$ **and** $\boldsymbol{\bar{D} = \varepsilon_0 \bar{E}}$, then above equation becomes

$$\mathbf{W_E = \frac{1}{2}\int_v (\bar{D} \cdot \bar{E}) dv = \frac{1}{2}\int_v \varepsilon_0 E^2 dv = \frac{1}{2}\int_v \frac{D^2}{\varepsilon_0} dv}$$

From this, we can define electrostatic energy density w_E (in J/m^3) as

$$\boldsymbol{w_E = \frac{dW_E}{dv} = \frac{1}{2}(\bar{D} \cdot \bar{E}) = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\frac{D^2}{\varepsilon_0}}$$

We can write

$$\mathbf{W_E = \int_v w_E dv}$$

<u>Current:</u>

The current through a given area is the electric charge passing through the area per unit time.

Or

The current is the rate of flow of charge

$$\mathbf{I = \frac{dQ}{dt} \ \ Amperes}$$

<u>Current density(J):</u>

Current density at a given point is the current passing through a unit normal area at that point. If current ΔI flows through a surface dS, the current density is

$$\mathbf{J = \frac{\Delta I}{\Delta S} \quad A/m^2}$$

$$\mathbf{\Delta I = J\Delta S}$$

$$\mathbf{I} = \int_{\mathbf{S}} \mathbf{J} \cdot \mathbf{dS}$$

CONVECTION CURRENT:

Convection current occurs when current flows through an insulating medium such as liquid, rarefied gas, or a vacuum. Convection current does not involve conductors and hence does not satisfy Ohm's law.

The Convention current density

$$\boxed{\mathbf{J} = \rho_v \, \mathbf{u}}$$

Conduction current :

Conduction current is defined as the current produced due to flow of electrons in conductors on the application of electric field.

- The conduction current density is given by

$$\boxed{\bar{\mathbf{J}} = \boldsymbol{\sigma}\bar{\mathbf{E}}}$$

- The above equation is called point form of ohm's law.

DIELECTRIC CONSTANT:

The dielectric constant (or relative permittivity) ε_r, is the ratio of the permittivity of the dielectric to that of free space.

$$\boxed{\boldsymbol{\varepsilon_r} = \frac{\boldsymbol{\varepsilon}}{\boldsymbol{\varepsilon_0}} = \mathbf{1} + \boldsymbol{\chi_e}}$$

$\boldsymbol{\chi_e}$ is the electric susceptibility

The **dielectric strength** is the maximum electric field that a dielectric can tolerate or withstand without breakdown.

Linear, Isotropic and homogeneous Dielectrics:

A dielectric material is linear if ε does not change with the applied electric field, isotropic if ε does not change with direction and homogeneous if ε does not change from point to point.

CONTINUITY EQUATION:

According to the principle of charge conservation, the time rate of decrease of charge within a given volume must be equal to the net outward current flow through the closed surface of the volume. Thus current I coming out of the closed surface is

$$\mathbf{I} = \oint \bar{\mathbf{J}} \cdot \mathbf{d}\bar{\mathbf{S}} = \frac{-\mathbf{dQ}}{\mathbf{dt}}$$

Where Q is the total charge enclosed by the closed surface.

According to divergence theorem

$$\oint_{\mathbf{s}} \bar{\mathbf{J}} \cdot \mathbf{d}\bar{\mathbf{S}} = \int_{\mathbf{v}} \boldsymbol{\nabla} \cdot \bar{\mathbf{J}} \mathbf{dv}$$

$$\frac{-\mathbf{dQ}}{\mathbf{dt}} = -\frac{\mathbf{d}\int_{\mathbf{v}} \boldsymbol{\rho}_{\mathbf{v}} \mathbf{dv}}{\mathbf{dt}} = -\int_{\mathbf{v}} \frac{\boldsymbol{\partial}\boldsymbol{\rho}_{\mathbf{v}}}{\boldsymbol{\partial}\mathbf{t}} \mathbf{dv}$$

$$\int_{\mathbf{v}} \boldsymbol{\nabla} \cdot \bar{\mathbf{J}} \mathbf{dv} = -\int_{\mathbf{v}} \frac{\boldsymbol{\partial}\boldsymbol{\rho}_{\mathbf{v}}}{\boldsymbol{\partial}\mathbf{t}} \mathbf{dv}$$

$$\boxed{\nabla \cdot \bar{J} = -\frac{\partial \rho_v}{\partial t}}$$

This equation is called the continuity of current equation.

For steady currents, $\frac{\partial \rho_v}{\partial t} = 0$ and hence

$$\boxed{\nabla \cdot \bar{J} = 0}$$

RELAXATION TIME:

Consider a linear and homogeneous medium with constants σ and ε for which current is flowing with current density

$$\bar{J} = \sigma \bar{E} = \frac{\sigma}{\varepsilon} \bar{D}$$

From continuity equation, we know that

$$\nabla \cdot \bar{J} = -\frac{\partial \rho_v}{\partial t}$$

$$\nabla \cdot \left(\frac{\sigma}{\varepsilon} \bar{D}\right) = -\frac{\partial \rho_v}{\partial t}$$

$$\frac{\sigma}{\varepsilon} \nabla \cdot \bar{D} = -\frac{\partial \rho_v}{\partial t}$$

From the Gauss's law , we know that

$$\nabla \cdot \bar{D} = \rho_v$$

$$\frac{\sigma}{\varepsilon} \rho_v = -\frac{\partial \rho_v}{\partial t}$$

$$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\varepsilon} \rho_v = 0$$

This is a first order differential equation in ρ_v whose solution is given by

$$\rho_v = \rho_{v0} e^{-t/T_r}$$

Where ρ_{v0} is the initial charge density.

The charge density inside the material is decays exponentially with time.

$$\boxed{T_r = \frac{\varepsilon}{\sigma}}$$

The time constant T_r is known as the relaxation time.

Relaxation time is the time it takes a charge placed in the interior of a material to decay to 36.8 percent of its initial value.

POISSON'S AND LAPLACE'S EQUATIONS:

Poisson's and Laplace's equations are easily derived from Gauss's law (for a linear medium)

$$\nabla \cdot \bar{D} = \rho_v$$

$$\nabla \cdot \varepsilon \bar{E} = \rho_v$$

But

$$\bar{E} = -\nabla V$$

$$\nabla \cdot \varepsilon(-\nabla V) = \rho_v$$

$$-\varepsilon \nabla^2 V = \rho_v$$

$$\boxed{\nabla^2 V = -\frac{\rho_v}{\varepsilon}}$$

When $\rho_v = 0$ (charge-free region), the above equation then becomes

$$\boxed{\nabla^2 V = 0}$$

$$\boxed{\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0}$$

$$\boxed{\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = 0}$$

$$\boxed{\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 V}{\partial \phi^2} = 0}$$

CAPACITANCE:

The capacitance C of the capacitor is defined as the ratio of the magnitude of the charge on one of the plates to the potential difference between them. that is,

$$\boxed{\mathbf{C = \frac{Q}{V} = \frac{\oint_S \bar{D} \cdot d\bar{s}}{-\int \bar{E} \cdot d\bar{l}}}}$$

Parallel-Plate Capacitor:

Consider the parallel-plate capacitor as shown in fig.

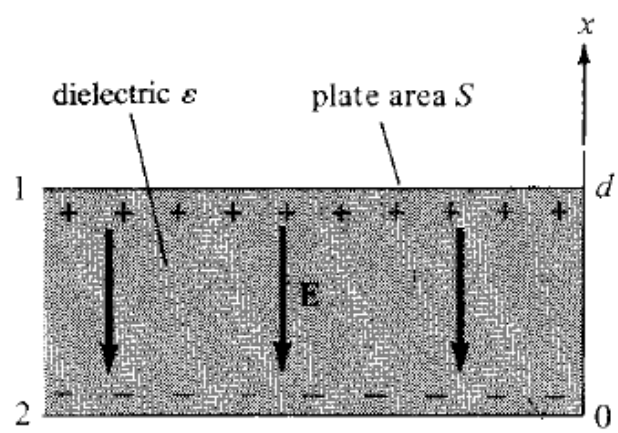


Fig. Parallel-plate capacitor

Let S be area of each plate and d be the distance between two plates. We assume that plates 1 and 2 carry charges +Q and -Q respectively so that

$$\boldsymbol{\rho_s = \frac{Q}{S}}$$

If the space between the plates is filled with a homogeneous dielectric with permittivity ε, then electric field intensity inside the plates is given by

$$\boldsymbol{\bar{E} = -\frac{\rho_s}{\varepsilon}\,\bar{a}_z}$$

$$\bar{E} = -\frac{Q}{\varepsilon S}\,\bar{a}_z$$

The potential difference between the plates is given by

$$V = -\int_2^1 \bar{E}\cdot d\bar{l}$$

$$V = -\int_0^d \left[-\frac{Q}{\varepsilon S}\,\bar{a}_z\right]\cdot dz\bar{a}_z$$

$$V = \int_0^d \frac{Q}{\varepsilon S}dz$$

$$V = \frac{Q}{\varepsilon S}\int_0^d dz$$

$$V = \frac{Qd}{\varepsilon S}$$

The capacitance for a parallel-plate capacitor is given by

$$C = \frac{Q}{V}$$

$$\boxed{C = \frac{\varepsilon S}{d}}$$

The energy stored in a capacitor is given by

$$\boxed{W_E = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{Q^2}{2C}}$$

<u>Coaxial Capacitor:</u>

Consider the Coaxial Cable as shown in fig.

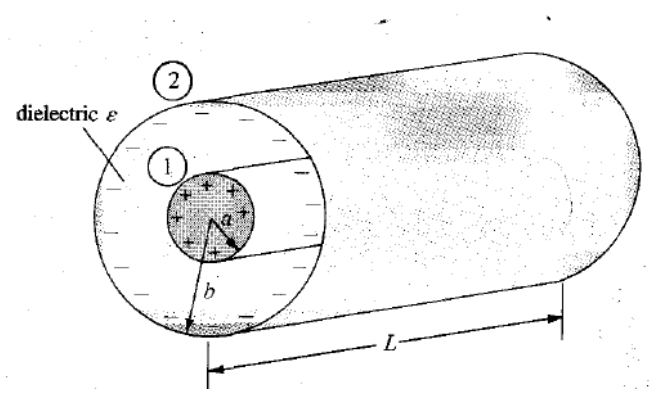


Fig. Coaxial capacitor.

Let a is the inner radius and b is the outer radius.

The length of the coaxial cable is L.

Let the space between the conductors be filled with a homogeneous dielectric with permittivity ε.

We assume that conductors 1 and 2 carry charges +Q and -Q respectively.

The electric field intensity due to infinite line charge is given by

$$\bar{E} = \frac{\rho_L}{2\pi\varepsilon\rho}\bar{a}_\rho$$

But

$$\rho_L = \frac{Q}{L}$$

$$\bar{E} = \frac{Q}{2\pi\varepsilon\rho L}\bar{a}_\rho$$

The potential difference V is given by

$$V = -\int_2^1 \bar{E}\cdot d\bar{l}$$

$$V = -\int_b^a \left[\frac{Q}{2\pi\varepsilon\rho L}\bar{a}_\rho\right]\cdot d\rho\bar{a}_\rho$$

$$V = -\int_b^a \frac{Q}{2\pi\varepsilon\rho L}d\rho$$

$$V = -\frac{Q}{2\pi\varepsilon L}\int_b^a \frac{d\rho}{\rho}$$

$$V = -\frac{Q}{2\pi\varepsilon L}[\ln\rho]_b^a$$

$$V = \frac{Q}{2\pi\varepsilon L}\ln\frac{b}{a}$$

Thus the capacitance of a coaxial cable is given by

$$C = \frac{Q}{V}$$

$$C = \frac{Q}{\frac{Q}{2\pi\varepsilon L}\ln\frac{b}{a}}$$

$$\boxed{C = \frac{2\pi\varepsilon L}{\ln\frac{b}{a}}}$$

<u>Spherical Capacitor:</u>

Consider a spherical capacitor as shown in fig.

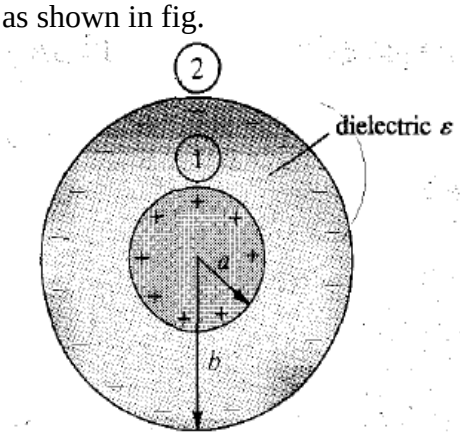


Fig. Spherical capacitor.

The inner radius of sphere is a and outer radius of sphere is b (b> a).Two spheres are separated by a dielectric medium with permittivity ε. We assume charges +Q and -Q on the inner and outer spheres respectively.
The electric field intensity between two concentric spheres is given by

$$\bar{E} = \frac{Q}{4\pi\varepsilon r^2}\,\bar{a}_r$$

The potential difference V between the spheres is given by

$$V = -\int_2^1 \bar{E}\cdot d\bar{l}$$

$$V = -\int_b^a \left[\frac{Q}{4\pi\varepsilon r^2}\,\bar{a}_r\right]\cdot dr\bar{a}_r$$

$$V = -\int_b^a \frac{Q}{4\pi\varepsilon r^2}dr$$

$$V = -\frac{Q}{4\pi\varepsilon}\int_b^a \frac{dr}{r^2}$$

$$V = -\frac{Q}{4\pi\varepsilon}\left[-\frac{1}{r}\right]_b^a$$

$$V = \frac{Q}{4\pi\varepsilon}\left[\frac{1}{a}-\frac{1}{b}\right]$$

Thus the capacitance of the spherical capacitor is

$$C = \frac{Q}{V}$$

$$C = \frac{Q}{\frac{Q}{4\pi\varepsilon}\left[\frac{1}{a}-\frac{1}{b}\right]}$$

$$\boxed{C = \frac{4\pi\varepsilon}{\left[\frac{1}{a}-\frac{1}{b}\right]}}$$

When b→∞ then

$$\boxed{C = 4\pi\varepsilon a}$$

Which is the capacitance of a spherical capacitor whose outer plate is infinitely large i.e isolated sphere.

<u>POLARIZATION IN DIELECTRICS:</u>

The dielectric is said to be polarized when a dipole results from the displacement of the charges on the application of electric field.
Dipole moment P=Qd.

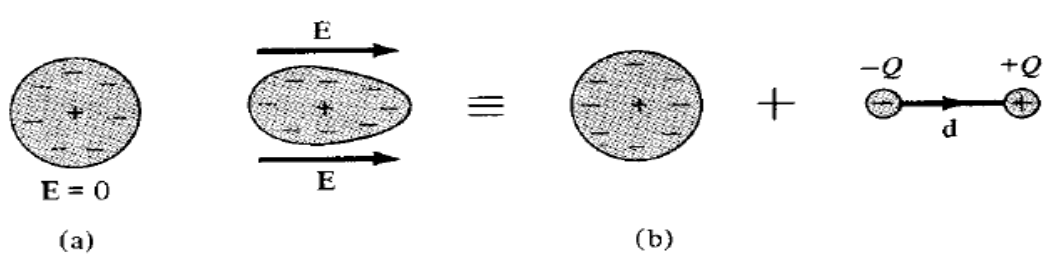


Fig. Polarization of a nonpolar dielectric atom.

- **Nonpolar dielectrics:**
- Nonpolar dielectric molecules do not possess dipoles until the application of the electric field.
- **Examples:** hydrogen, oxygen, nitrogen, and the rare gases.
- **polar dielectrics:**
- The polar dielectric molecules have built-in permanent dipoles that are randomly oriented.
- **Examples:** water, sulfur dioxide, and hydrochloric acid

- **Properties of dielectric materials:**
- The dielectrics do not contain any free charges but contain bounded charges.
- Bounded charges are under the influence of internal forces and can't contribute to the conduction.
- The dielectrics are polarized under the application of electric field.
- The dielectrics store energy due to polarization.